

EXPERIMENT7

Name: Bhavik Pahuja
Branch: B.E. CSE (AIML)
Semester: 5
Subject Name: ADBMS

UID: 24BAI70522
Section: 24AIT_KRG-1_B
Date of Performance: 31.08.2026
Subject Code: 24CSP-310

1. Consider a relation R having attributes as R(ABCD), functional dependencies are given below:
 $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Solution:

$B^+ = B$

$AB^+ = ABCD$

$BC^+ = BCDA$

$BD^+ = BDAC$

Candidate Keys: AB, BC, BD

Prime Attributes: {A, B, C, D}

Non-Prime Attributes: {}

Since every FD either has a superkey as its determinant or has a prime attribute on its RHS, the relation is in 3NF. It is not in BCNF because C and D are not superkeys in $C \rightarrow D$ and $D \rightarrow A$.
Highest Normal Form: 3NF

2. Relation R(ABCDE) having functional dependencies as :

$A \rightarrow D$, $B \rightarrow A$, $BC \rightarrow D$, $AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Solution:

$C^+ = C$

$AC^+ = ACDBE$

$BC^+ = BCAD E$

$CD^+ = CD$

$CE^+ = CE$

Candidate Keys: AC, BC

Prime Attributes: {C, A, B}

Non-Prime Attributes: {D, E}

AC and BC are candidate keys, and since A (part of AC) determines non-prime attribute D, there is a partial dependency. Hence, the relation is in 1NF. Highest Normal Form: 1NF

3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:
 $B \rightarrow A$, $A \rightarrow C$, $BC \rightarrow D$, $AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Solution:

$A^+ = ACBED$

$B^+ = ABCDE$

$BC^+ = BCDAE$

$AC^+ = ACBED$

CK: {A, B}

PA: {A, B, C} NPA:
{D, E}

Since all determinants are super keys, the table is in BCNF.

Highest Normal Form: BCNF

4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:

$A \rightarrow BCD$, $BC \rightarrow DE$, $B \rightarrow D$, $D \rightarrow A$

Identify the set of candidate keys possible in relation R. List all the set of prime and non prime attributes.

Solution:

$F^+ = \{F\}$

$AF^+ = ABCDEF$

$BF^+ = ABCDEF$

$DF^+ = ABCDEF$

CK: {AF, BF, DF}

PA: {A, B, D, F}

NPA: {C, E}

The relation is not in 2NF because A, which is a part of the candidate key AF, partially determines the non-prime attribute C ($A \rightarrow C$). Highest Normal Form: 1NF

5. Relation R(ABCDE) having dependencies as:

$CE \rightarrow D$

$D \rightarrow B$

$C \rightarrow A$

Highest normal form? C.K set?

Solution:

$CE^+ = CEDBA$

CK: {CE}

PA: {C, E}

NPA: {A, B, D}

Since C, which is part of the candidate key CE, determines the non-prime attribute A ($C \rightarrow A$), there is a partial dependency. Hence, the relation is in 1NF. Highest Normal Form: 1NF

6. Relation R(ABCDEF) having dependencies as:

$AB \rightarrow C$

$DC \rightarrow AE$

$E \rightarrow F$

Highest normal form? C.K set?

Solution:

$BD^+ = BD$

$ABD^+ = ABCDEF$

$BCD^+ = ABCDEF$

CK: {ABD, BCD}

PA: {A, B, C, D}

NPA: {E, F}

Since DC is a proper subset of candidate key BCD and determines the non-prime attribute E, a partial dependency exists.

Highest Normal Form: 1NF

7. Relation R(ABCDEFGHI) having dependencies as:

$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

Highest normal form? C.K set?

Solution:

$ABD^+ = ABCDEFGHI$

CK: {ABD}

PA: {A, B, D}

NPA: {C, E, F, G, H, I}

Since A, which is part of candidate key ABD, determines non-prime attribute I ($A \rightarrow I$), there is a partial dependency.

Highest Normal Form: 1NF

8. Relation R(ABCDE) having dependencies as:

$AB \rightarrow CD$

$D \rightarrow A$

$BC \rightarrow DE$

Highest normal form? C.K set?

Solution:

$AB^+ = ABCDE$

$BC^+ = BCDEA$

$BD^+ = BDACE$

CK: {AB, BC, BD}

PA: {A, B, C, D}

NPA: {E}

There is no partial dependency, and for $D \rightarrow A$, A is a prime attribute, thus, the relation satisfies 3NF.

Highest Normal Form: 3NF

9. Relation R(ABCDE) having dependencies as:

$BC \rightarrow ADE$

$D \rightarrow B$

Highest normal form? C.K set?

Solution:

$C^+ = C$

$AC^+ = AC$

$BC^+ = ADEBC$

$CD^+ = CDBAE$

CK: {BC, CD}

PA: {B, C, D}

NPA: {A, E}

There is no partial dependency involving a non-prime attribute, and in $D \rightarrow B$, B is a prime attribute, therefore, the relation satisfies 3NF. However, D is not a superkey, so it is not in BCNF.

Highest Normal Form: 3NF

10. Relation R(ABC) having dependencies as:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow A$

Highest normal form? C.K set?

Solution:

$A^+ = ABC$

$B^+ = BCAC^+$

$= CAB$

CK: {A, B, C}

PA: {A, B, C} NPA:

{}

Since the determinant in every FD is a candidate key, the relation is in BCNF. Highest Normal Form: BCNF

11. Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

$F1 \rightarrow F3$

$F2 \rightarrow F4$

$(F1, F2) \rightarrow F5$

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Solution:

$(F1 F2)^+ = F1 F2 F3 F4 F5$

CK: {F1 F2}

PA: {F3, F4}

NPA: {F3, F4, F5}

Since F1 and F2, which are parts of the candidate key, determine non-prime attributes, partial dependency exists.

Highest Normal Form: 1NF

12. Consider a relation R(ABCDEFGHI) having FDs as:

$AB \rightarrow C$

$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

$H \rightarrow J$

☐ Identify that relation is in which normal form and the candidate key by determining the prime and non prime attributes.

- ☐ Apply the decomposition if needed to remove the redundancy.
- ☐ Identify how many new tables will be formed after the decomposition.

Solution:

$ABD^+ = ABCDEFGHIJ$

CK: {ABD}

PA: {A, B, D}

NPA: {C, E, F, G, H, I, J}

Since parts of the candidate key determine non-prime attributes, partial dependency exists. Highest Normal Form: 1NF

Decomposition:

R1(ABC)

R2(ADGH)

R3(BDEF)

R4(AI)

R5(HJ)

R6(ABD)

Number of tables formed: 6

13. Relation R having attributes as (A, B, C, D, E) having the functional dependencies as follows:

$A \rightarrow B$

$B \rightarrow E$

$C \rightarrow D$

- ☐ Identify that relation is in which normal form and the candidate key by determining the prime and non prime attributes.
- ☐ Apply the decomposition if needed to remove the redundancy.
- ☐ Identify how many new tables will be formed after the decomposition.

Solution:

$AC^+ = ACBED$

CK: {AC}

PA: {A, C}

NPA: {B, D, E}

Since A and C, which are parts of candidate key AC, determine non-prime attributes, partial dependency exists. Highest Normal Form: 1NF

Decomposition:

R1(AB)

R2(BE)

R3(CD)

R4(AC)

Number of tables formed: 4